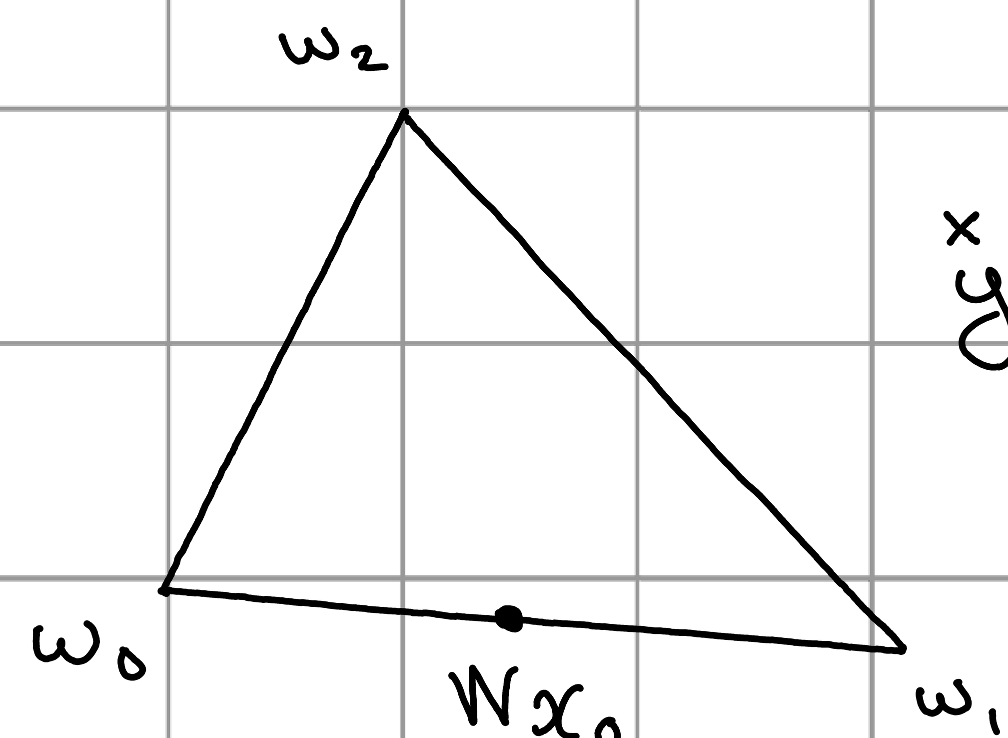
Problem : $\min_{x \geq 0} \left\| \underset{\text{normalized}}{\overset{\in \mathbb{R}_+^3}{y}} - [w_0, w_1, w_2]\, x \right\|_2^2$

normalized

Initialization : $S_0 = \{0, 1\}$, $x_0 = [*, *, 0]$

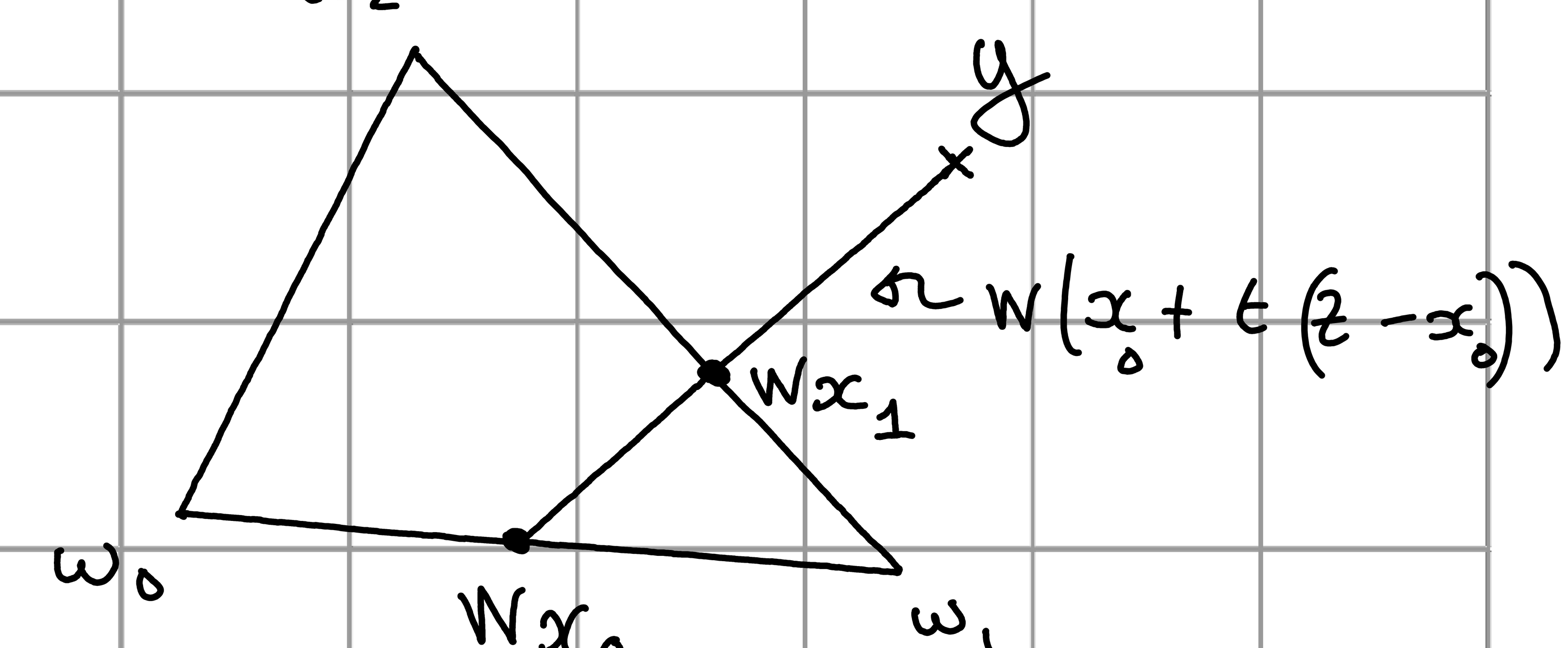
First Step : add $\{2\}$ to S_0 ; $S_1 = \{0, 1, 2\}$

Second Step : $z = W^\dagger y$, $Wz = y$



$\overset{\times}{y} = Wz$

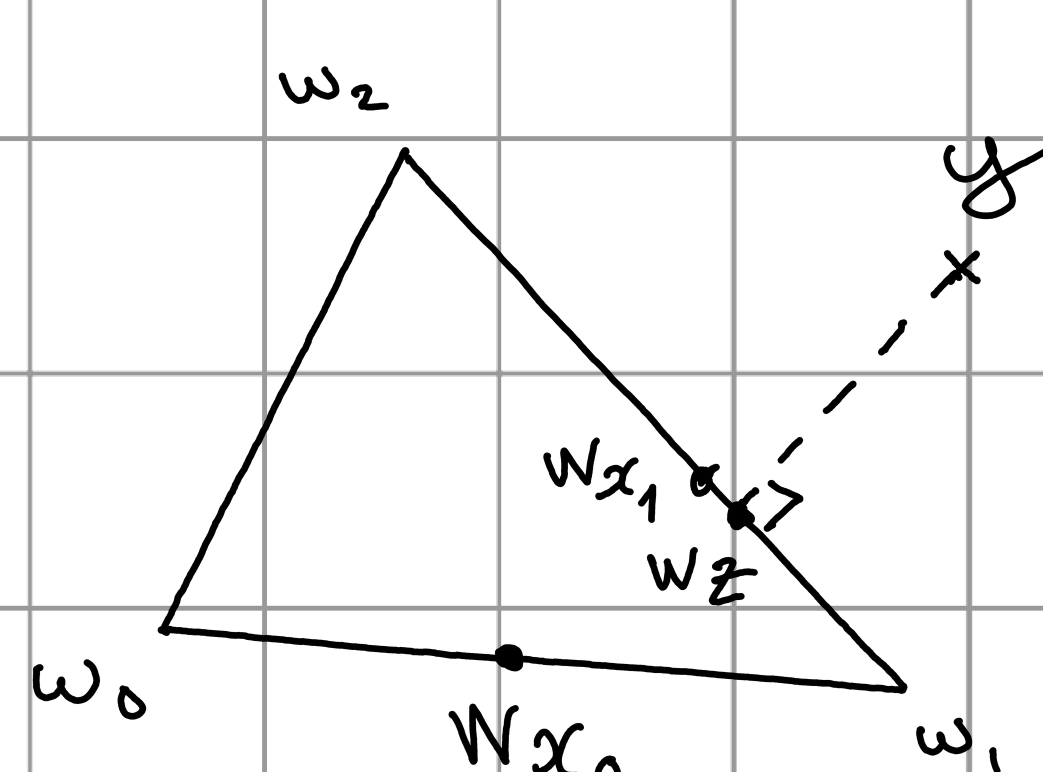
z contains negative entries



Third Step : $t^* = \dfrac{x_0[0]}{x_0[0] - z[0]}$

$x_1 = x_0 + t^*(z - x_0)$

Second Step (2d iteration) : $z = [w_1, w_2]^\dagger y \geq 0$



Wz is the orthogonal projection of y on the cone $\langle W \rangle^+$